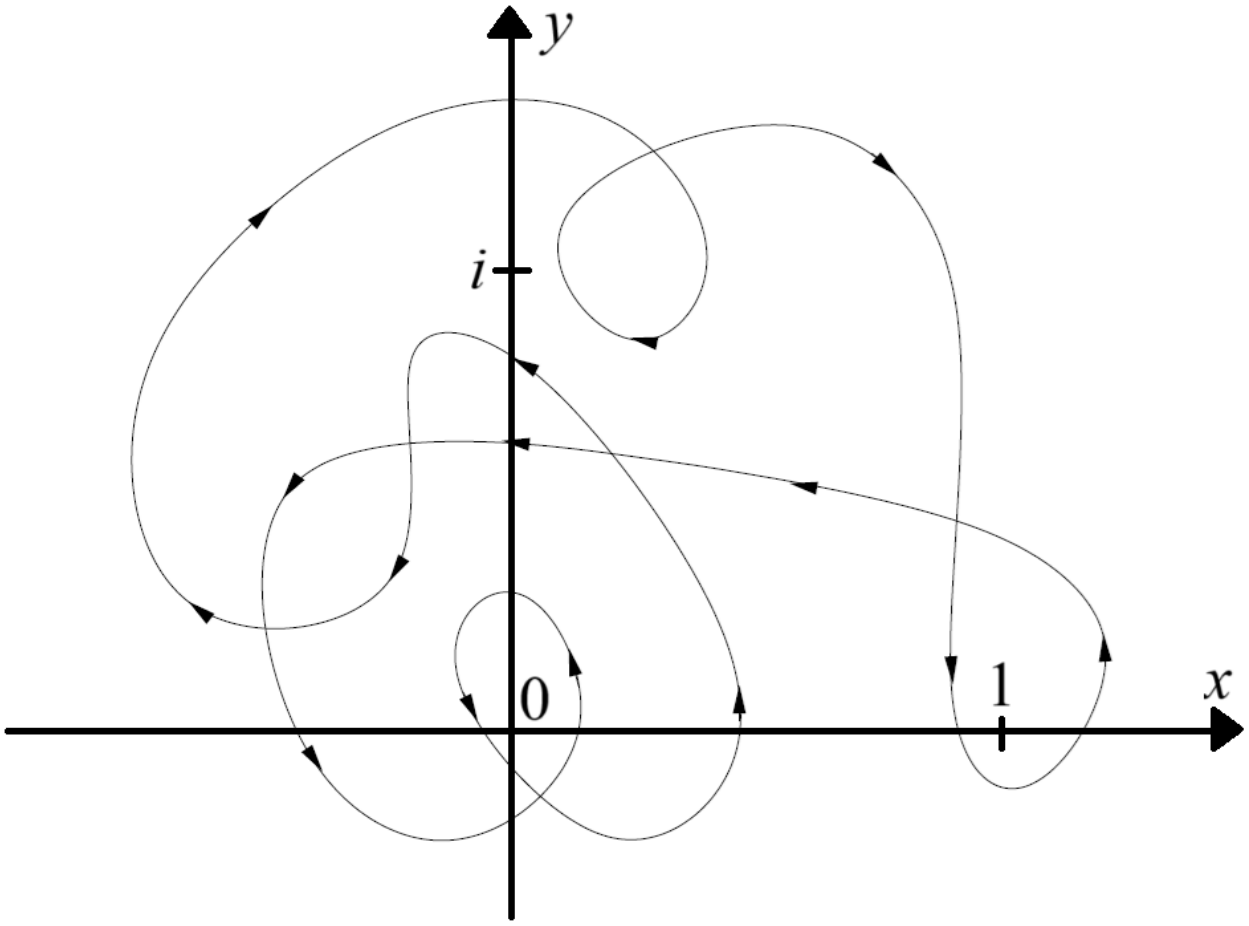
Vector Calculus and Complex Analysis Classwork (2025/4/28)

Name:

Student ID:

Let C be the following curve.



Calculate $\int_C \frac{1}{z(z-1)(z-i)}\,dz$.

Solution

$$\frac{1}{z(z-1)(z-i)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-i}$$

$$A = \lim_{z\to 0} \frac{1}{(z-1)(z-i)} = \frac{1}{i} = -i$$

$$B = \lim_{z\to 1} \frac{1}{z(z-i)} = \frac{1}{1-i} = \frac{1}{2}(1+i)$$

$$C = \lim_{z\to i} \frac{1}{z(z-1)} = \frac{1}{i(i-1)} = \frac{1}{-1-i} = -\frac{1}{2}(1-i)$$

$$\int_C \frac{1}{z(z-1)(z-i)}\,dz = 2\pi i\left(2A + B - C\right) = 2\pi i\left(-2i + \frac{1}{2}(1+i) + \frac{1}{2}(1-i)\right) = 2\pi i(1-2i) = 2\pi(2+i)$$